



UNIVERSITY OF THE PACIFIC

EBERHARDT SCHOOL OF BUSINESS

Module 1

Practical Homework Assignment

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MSBA 265: Business Analytics Topics Foundational

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Part 1: Workspace Setup, Git Ingestion, & Procurement

The Module 1 project workspace was configured in Visual Studio Code using a Python virtual environment (venv) to isolate project dependencies. Git version control was initialized, and a .gitignore file was configured to exclude the virtual environment, Python cache files, and Jupyter checkpoint files. The project directory was organized into separate folders for data, notebooks, source scripts, reports, and generated figures to support a clear and reproducible workflow.

The data/download_data.py script was created and executed to programmatically retrieve the French Motor Third Party Liability Claims dataset (freMTPL2freq) from OpenML. The downloaded dataset was saved as data/raw_business_data.csv and contained **678,013 records and 12 features**, confirming successful data procurement.

After the workspace and dataset were prepared, the project files were committed using Git and pushed to a public GitHub repository. The repository includes the required source code, notebook, dependencies, generated artifacts, final report, and README instructions.

Part 2: Raw Boundary Audits & Business Data Dictionary

2.1 Structural Audit and Summary Statistics

A structural audit was performed using df.info() to examine the dataset dimensions, column names, data types, and missing values.

The command df.describe().T was also used to generate descriptive statistics for the numerical features, including count, mean, standard deviation, minimum, quartiles, and maximum values.

The dataset contains **678,013 policy records across 12 features**. The structural audit confirmed that the analyzed fields contained no missing values requiring imputation.

2.2 Raw Boundary Audit

Minimum and maximum values were reviewed for DrivAge, Exposure, and BonusMalus to identify possible invalid values, negative values, or sentinel null codes.

The raw boundary audit produced the following results:

- **DrivAge:** Minimum = 18, Maximum = 100
- **Exposure:** Minimum = 0.0027, Maximum = 2.01
- **BonusMalus:** Minimum = 50, Maximum = 230

No negative values or obvious sentinel null codes were identified. DrivAge remained within valid adult driving ages, Exposure contained no negative policy durations, and BonusMalus remained within its expected regulatory range.

These results indicate that the reviewed variables do not contain obvious boundary-related data quality issues.

2.3 Business Data Dictionary

A Business Data Dictionary was generated to document the analytical and business meaning of each feature.

The dictionary includes:

- Feature name
- Business label
- Data type
- Unit
- Null count
- Number of unique values
- Business definition

The completed data dictionary was exported as: [reports/data_dictionary.csv](#)

The Business Data Dictionary provides the domain context necessary to correctly interpret each variable before applying transformations or modeling techniques.

2.4 Exposure Offset and Density/Area Redundancy

Exposure acts as a rate offset because the number of claims depends on how long a policy is active. For example, a policy that is active for one year has more opportunity to generate claims than a policy that is active for only six months, so claim counts should be adjusted based on Exposure. Density and Area should not both be included in the same linear regression model because they represent similar information about population density. Density gives a continuous numerical value, while Area groups population density into categories. Including both variables would add redundant information to the model and could create collinearity problems.

Part 3: Overriding Automated Rules & Correlation Visuals

3.1 Pearson Correlation Heatmap



A Pearson Correlation Heatmap was generated to evaluate the strength and direction of linear relationships among the numerical features.

- **Distribution/Relationship Shape:** The heatmap shows mostly weak linear relationships between the numerical variables, with most Pearson correlation values close to 0.
- **Key Metrics:** The strongest visible relationship is between DrivAge and BonusMalus at approximately $r = -0.48$, indicating a moderate negative relationship. No numerical feature pair reaches the severe multicollinearity threshold of $|r| \geq 0.85$.
- **Recommended Analytics Action:** No numerical feature needs to be removed based only on the Pearson correlation heatmap. However, Density and Area should still not both be used because they represent similar geographic information, even though Area is categorical and therefore does not appear in the numerical correlation heatmap.

3.2 Override of Automated Skewness Recommendation for ClaimNb.

```
=====
UNIVARIATE SKEWNESS DIAGNOSTICS
=====
```

	Feature	Skewness Coefficient	Diagnosis	Action Required
0	ClaimNb	5.599613	Extremely Right-Skewed	Log/Box-Cox Transform
1	Density	4.651421	Extremely Right-Skewed	Log/Box-Cox Transform
2	BonusMalus	1.728934	Moderately Skewed	Log/Box-Cox Transform
3	VehPower	1.171344	Moderately Skewed	Log/Box-Cox Transform
4	VehAge	1.147989	Moderately Skewed	Log/Box-Cox Transform
5	DrivAge	0.435759	Unskewed/Symmetric	Standard Scaler Sufficient
6	IDpol	0.237890	Unskewed/Symmetric	Standard Scaler Sufficient
7	Exposure	0.085318	Unskewed/Symmetric	Standard Scaler Sufficient

The automated skewness analysis identifies ClaimNb as extremely right-skewed, with a skewness coefficient of approximately 5.60, and recommends a log transformation. However, I would override this recommendation because ClaimNb is a discrete count variable, not a continuous positive variable. It can also contain zero values, and $\log(0)$ is mathematically undefined and approaches $-\infty$. Therefore, ClaimNb should remain in its original count form and be modeled using a Poisson or Negative Binomial model instead of applying a log transformation.

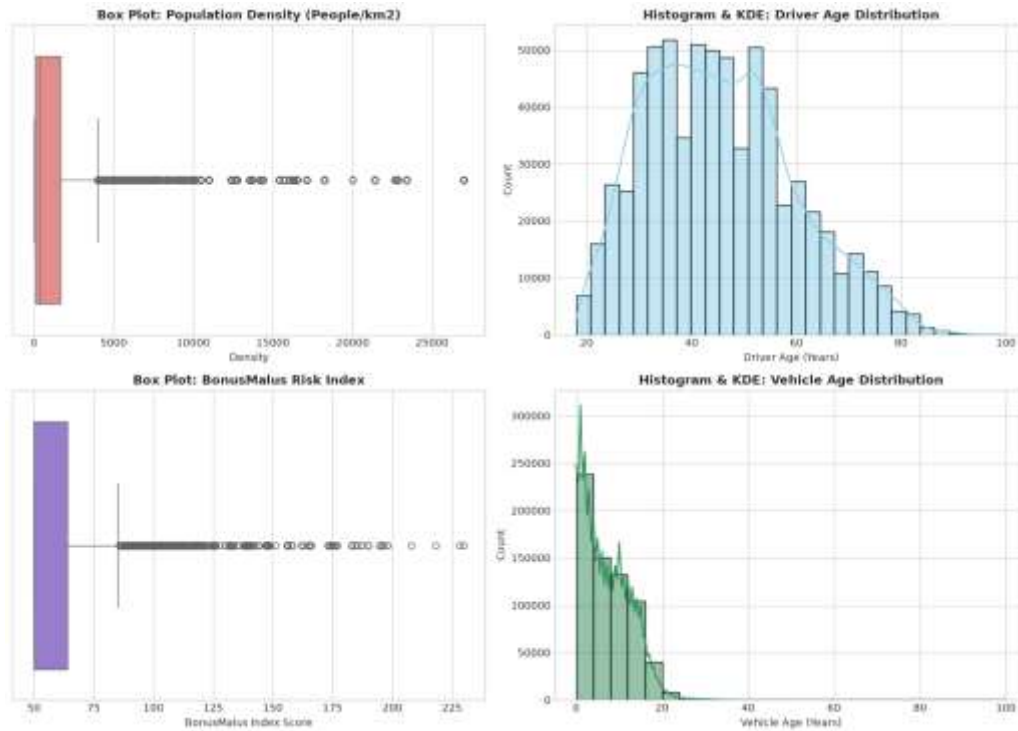
Part 4: Distribution Audits & Written Plot Statements

4.1 Render side-by-side box plots and KDE histograms

Four distribution plots were generated:

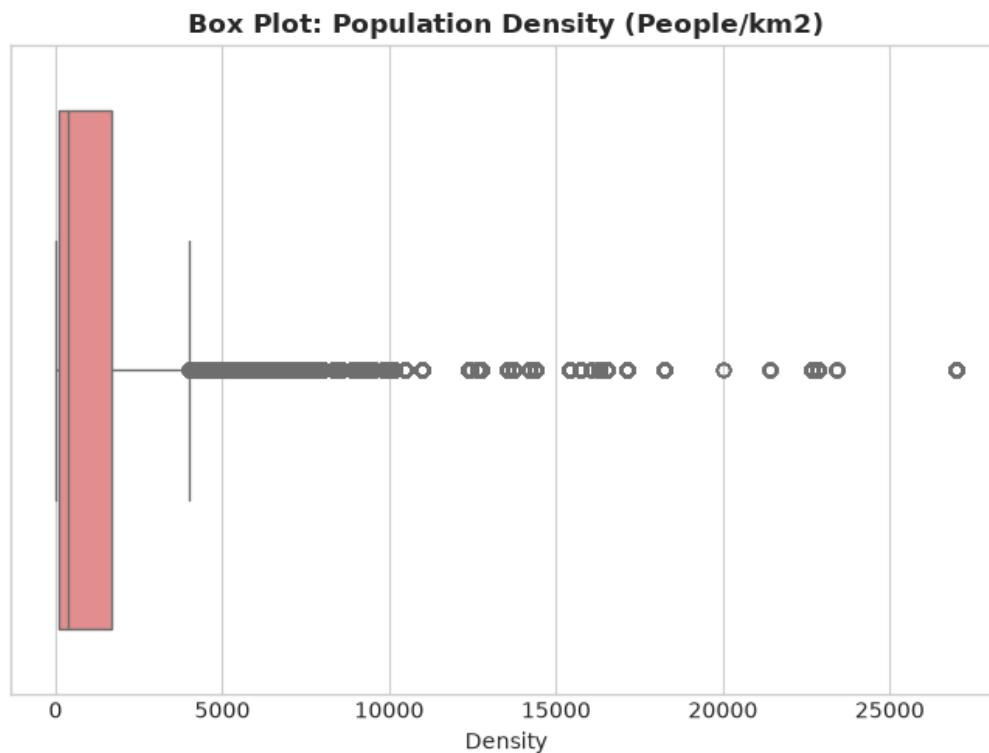
- Population Density Box Plot
- Driver Age Histogram and KDE
- BonusMalus Risk Index Box Plot
- Vehicle Age Histogram and KDE

The complete figure was exported as: [reports/figures/feature_distributions.png](#)



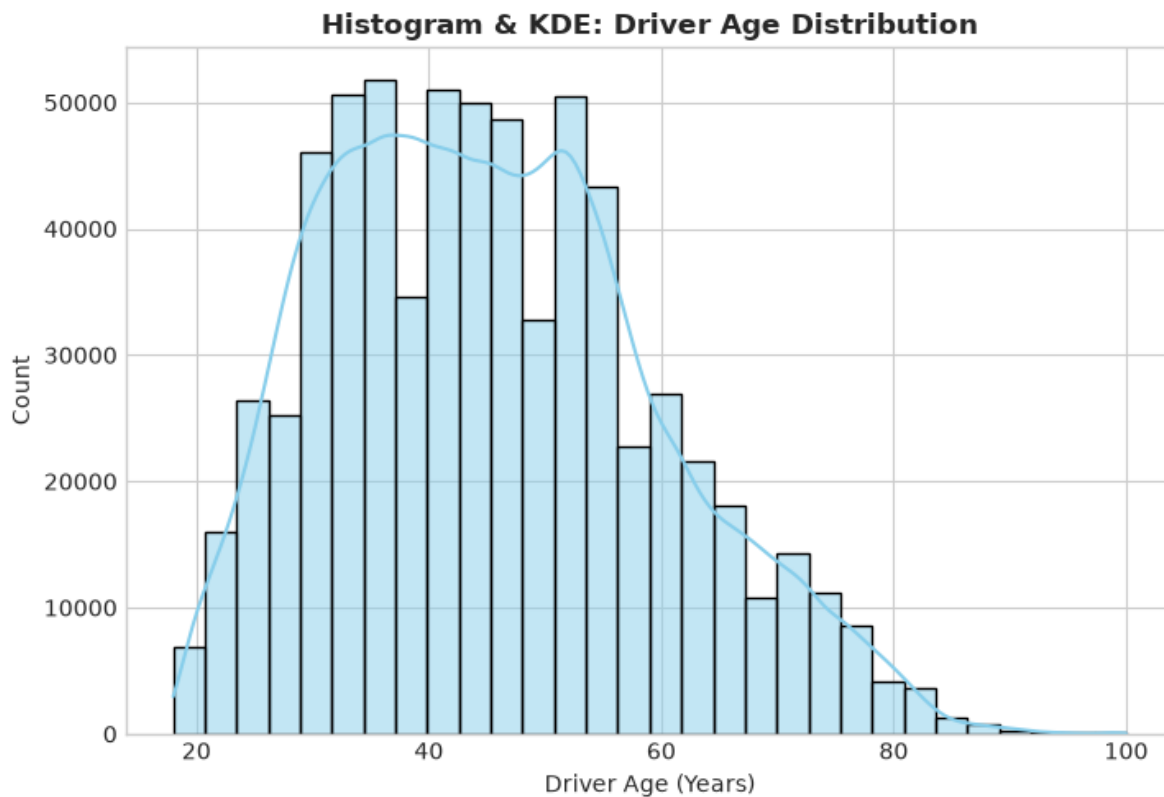
4.2 Write the required 3-bullet interpretation statements (Distribution Shape, Key Metrics/Whiskers, Recommended Analytics Action)

4.2.1. Box Plot: Population Density (People/km2)



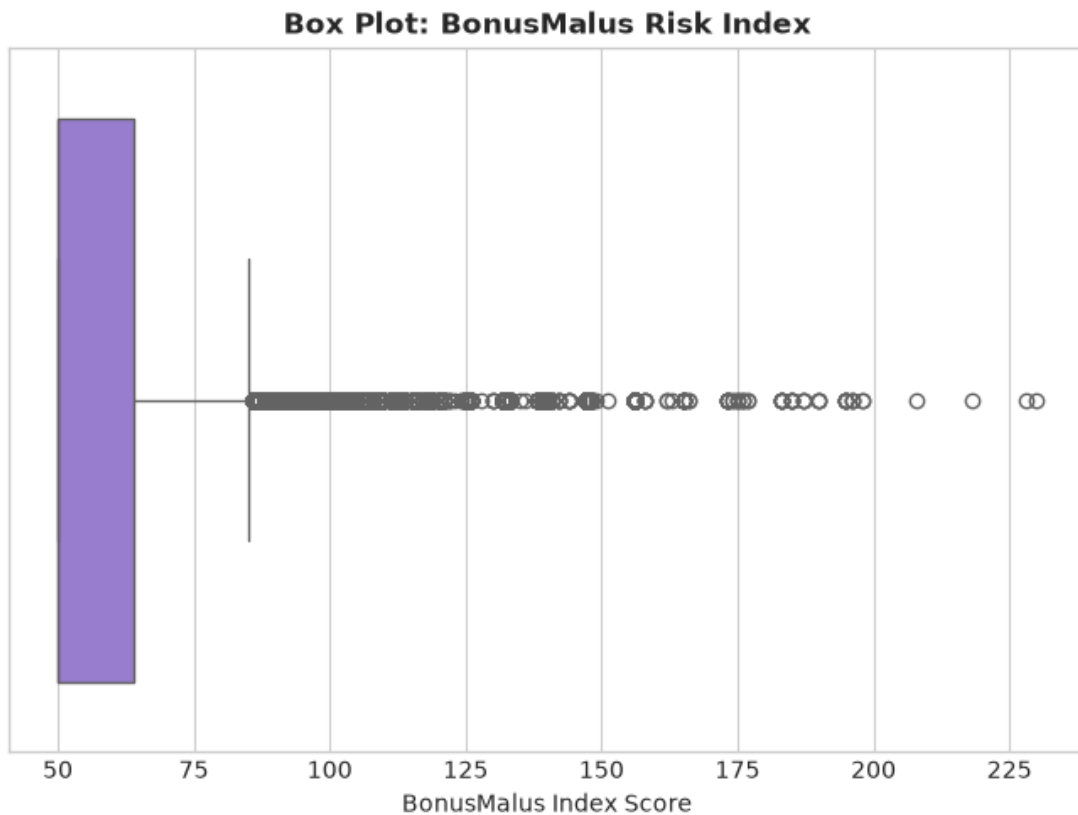
- **Distribution Shape:** Density is extremely right-skewed, with a skewness coefficient of approximately 4.65. Most observations are concentrated at lower population densities, while a long tail extends to the right.
- **Key Metrics/Whiskers:** The upper whisker is around 4,000 people/km², while many observations extend far beyond the whisker, with some values above 20,000 people/km². These points are statistical outliers but may represent valid high-density urban areas.
- **Recommended Analytics Action:** Keep the valid high-density observations and apply a $\log(1 + \text{Density})$ transformation to reduce the strong right skew before modeling.

4.2.2. Histogram & KDE: Driver Age Distribution



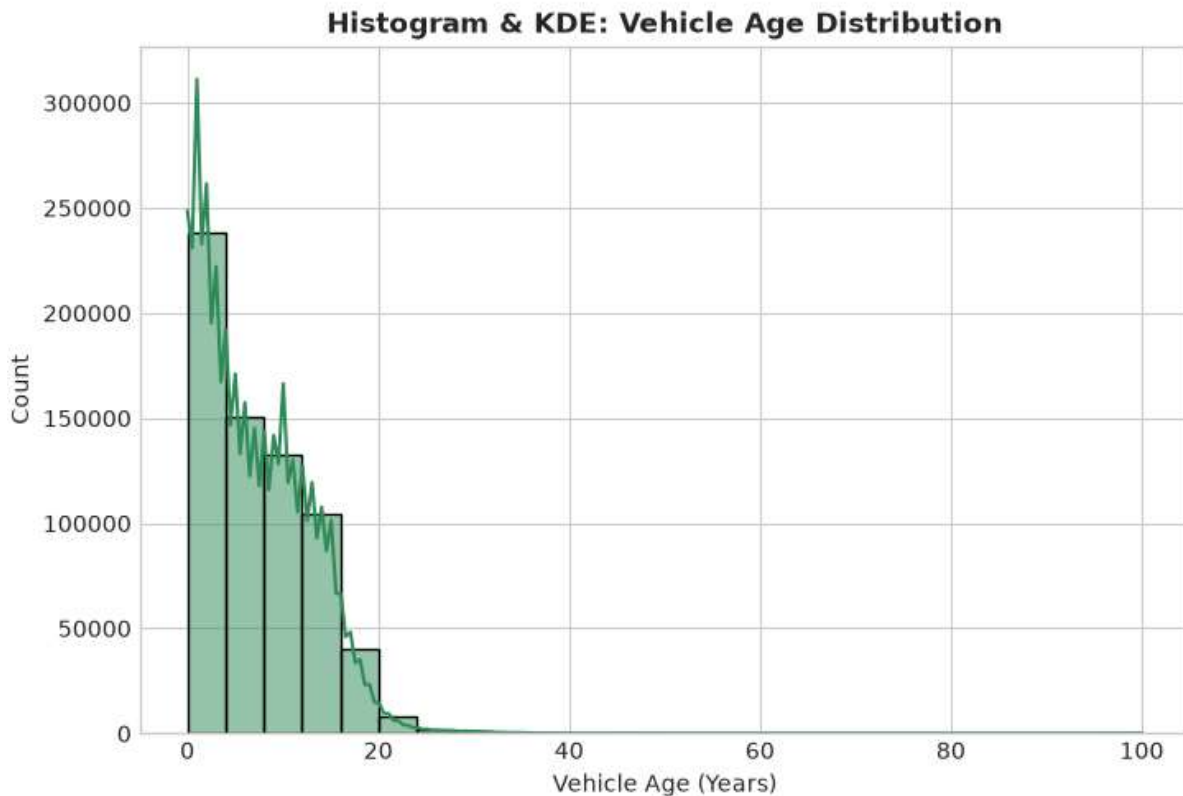
- **Distribution Shape:** DrivAge is relatively symmetric and close to a bell-shaped distribution, with a skewness coefficient of approximately 0.44.
- **Key Metrics/Whiskers:** Driver ages range from 18 to 100, and most drivers are concentrated between roughly 30 and 60 years old.
- **Recommended Analytics Action:** No log transformation or outlier trimming is needed. Standard Z-score scaling can be used before modeling.

4.2.3. Box Plot: BonusMalus Risk Index



- **Distribution Shape:** BonusMalus is moderately right-skewed, with a skewness coefficient of approximately 1.73, with most values concentrated at the lower end of the distribution.
- **Key Metrics/Whiskers:** Most scores are concentrated below 100, while values above the upper whisker appear as outliers and extend to approximately 230.
- **Recommended Analytics Action:** These higher values represent valid high-risk drivers rather than data errors, so they should be retained. BonusMalus can be scaled, and a binary indicator such as `Is_Surcharge_Driver = BonusMalus > 100` can also be created.

4.2.4. Histogram & KDE: Vehicle Age Distribution



- **Distribution Shape:** VehAge is moderately right-skewed, with a skewness coefficient of approximately 1.15. Most vehicles are concentrated at younger ages, with a long tail extending to the right.
- **Key Metrics/Whiskers:** Most vehicles are less than about 20 years old, while only a small number of observations extend to much older vehicle ages.
- **Recommended Analytics Action:** VehAge can be grouped into age categories or scaled before modeling to reduce the effect of the right-skewed distribution.

Part 5: Production Pipeline Scripting

A production script named [src/clean_outliers.py](#) was created to automate outlier filtering for the Density feature.

The script calculates:

- First quartile (Q1)
- Third quartile (Q3)
- Interquartile Range (IQR)

- Lower Tukey boundary
- Upper Tukey boundary

The Tukey **1.5 × IQR** method was then used to filter statistical outliers.

The script was executed using:

```
python src/clean_outliers.py
```

```
(venv) PS C:\Users\ngoho\OneDrive\Desktop\UOP\Fall 2026\MSBA 265\HW\msba265-module1-AnhNgo> python src/clean_outliers.py
=====
PRODUCTION OUTLIER FILTERING REPORT
=====
Target Feature Filtered: Density
Initial Dataset Records: 678,013
Tukey IQR Valid Range: [-2,257.00, 4,007.00]
Outlier Records Removed: 77,566 (11.44%)
Final Cleaned Records: 600,447
```

The original dataset contained **678,013 records**. A total of **77,566 records**, representing approximately **11.44%** of the dataset, were identified outside the Tukey IQR boundaries and removed.

The resulting cleaned dataset contains: **600,447 records**

The cleaned dataset was saved as: [data/cleaned_business_data.csv](#)

This production script provides a reproducible filtering process and ensures that the same outlier-cleaning logic can be applied consistently.